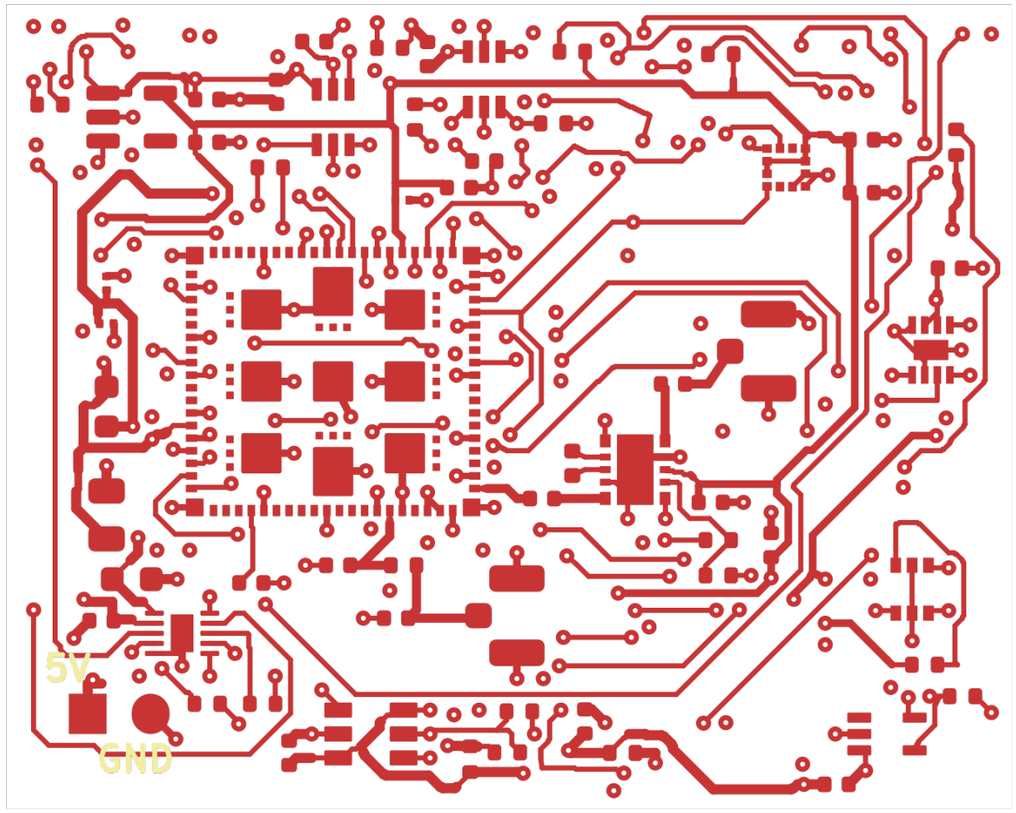
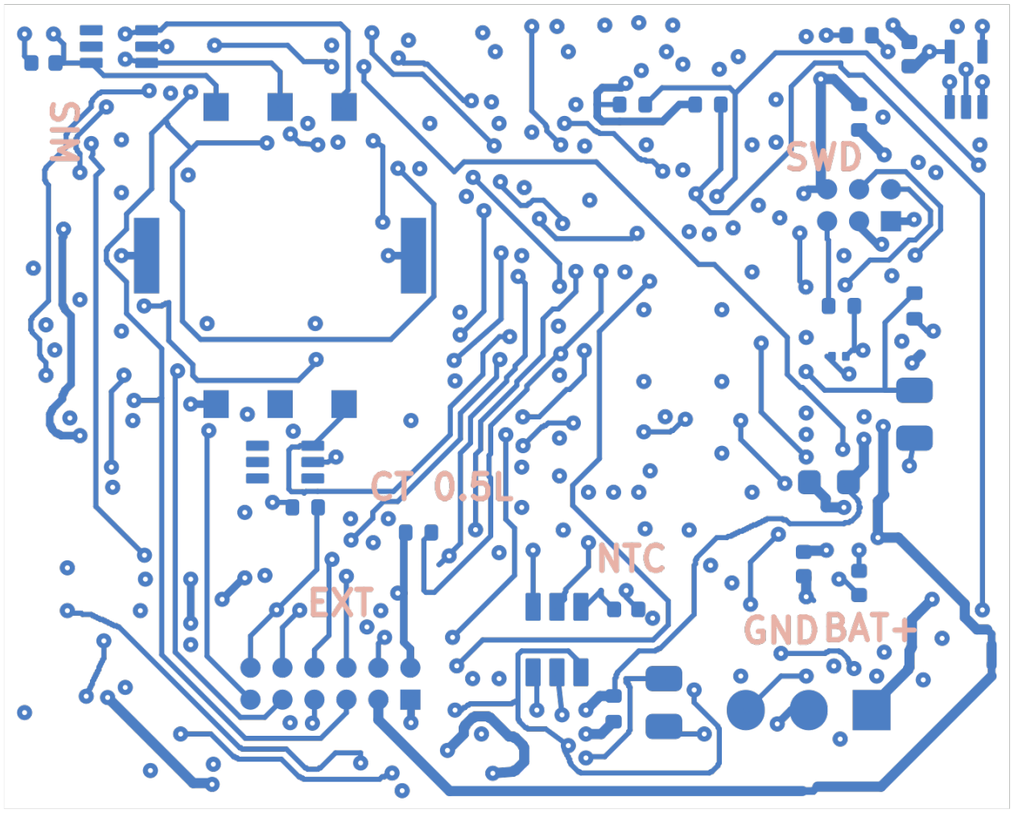
COLLAR TRACKER / r05l PCB REVIEW

Two actual all-green PCB DFM reports. Working-device release remains on hold.

TOP / copper and silkscreen



BOTTOM / mirrored component view



VERIFIED PCB / UNCHANGED

40 x 32 mm / six Cu layers / 87 footprints.

Native DRC, ERC, connectivity and parity: zero findings.

JLC: 0 Danger / 0 Warning / 455 Good in both runs.

Same ZIP, unchanged rules, all 29 rows; SMT off.

81 fitted references / 44 exact candidate MPNs.

109 physical GND pads have local via paths.

Two GND planes; 154/154 signal-via return screens.

REQUIRED BEFORE WORKING DELIVERY

Hardware charge-temperature interlock is implemented.

Its thermal/startup model still needs powered tests.

Protected pack and NTC attachment are unselected.

r08 prototype antenna/cable model: see next page.

Firmware, factory programming and functional tests: open.

Procurement, CPL rotations and SMT acceptance: open.

This package is for review, not an order release.

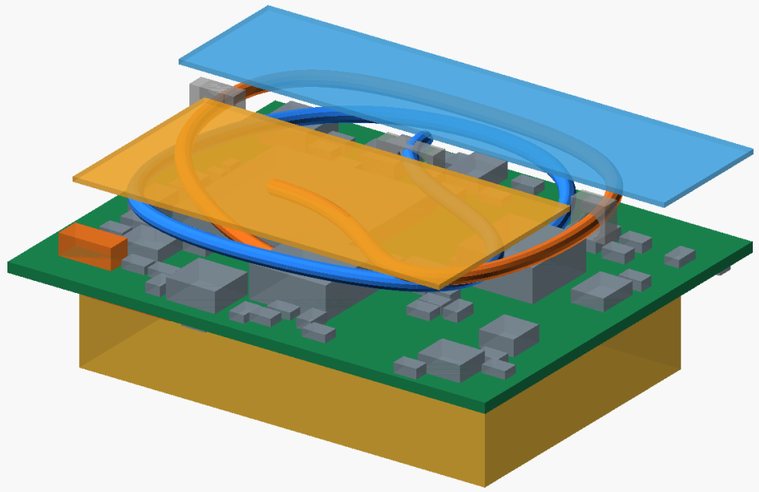
PCB SHA256: 241c84d6ba98793c37344e1e46ddc4918e7b05322ba87daea7c9a90ffb8379b3

JLC records: 637366592129687553 / 637366801595977730

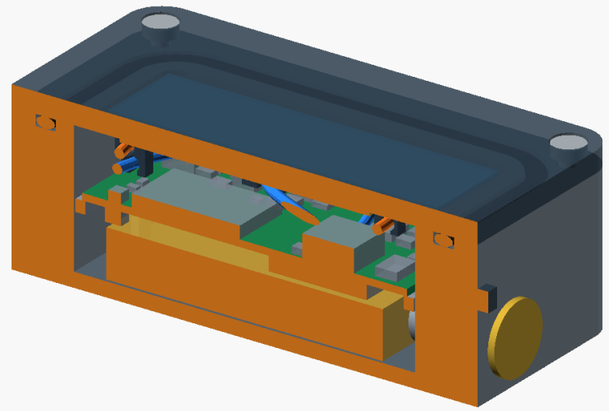
COLLAR TRACKER / r08 ASSEMBLY REVIEW

Prototype antennas and routed cables. The all-green r05l PCB is unchanged.

LTE + GNSS / actual connector positions



SECTION / seal, carrier and cable guides



CHECKED MODEL / CONDITIONAL

CAPH LTE + ZETA GNSS; standard 100 mm cable candidates.

85 populated envelopes; four PCB thickness cases.

Separate cable paths and four internal lid guides.

CAD guide interference empty; undersized control fails.

Eight closed meshes; five inspected assembly views.

Rigid shell + lid: 57.4 x 49.4 x 20.6 mm.

Boot: 60.4 x 52.4 x 25.4 mm. Key adds 1.3 mm.

WORKING-DEVICE RELEASE STILL ON HOLD

Protected pack / NTC attachment: not selected.

Cable end datums and length tolerance fit: open.

Exact coax bend limit and feed bond: open.

Clip strength and opening/service motion: untested.

PCB retention, dock/harness and factory firmware: open.

No powered, RF or physical water/dust test exists.

IP68 is a design target, not an achieved rating.

PCB SHA256: 241c84d6ba98793c37344e1e46ddc4918e7b05322ba87daea7c9a90ffb8379b3

Two preserved JLC runs: 0 Danger / 0 Warning / 455 Good each. SMT was not analyzed.